

Do Not Anonymize Away the Answer: Measuring Task-Critical Meaning Loss in Text De-identification

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Abstract

Text de-identification is often evaluated as a span removal problem: did a system remove names, dates, locations, contact information, and other explicit identifiers? For sensitive text used in foundation-model workflows, this is necessary but incomplete. A transformed clinical vignette or legal case excerpt can pass a privacy-oriented screen while removing the medication dose, lab value, statutory article, procedural posture, or legal claim needed for downstream reasoning. We study this failure mode through a small clinical/legal benchmark for utility-preserving anonymization. The benchmark pairs privacy labels with task-critical facts and downstream QA items. We introduce *Task-Critical Fact Retention* (TCFR), a fact-level utility metric, and Critical Span Guard, a low-cost protocol that extracts privacy spans and important domain facts, anonymizes under preservation constraints, verifies the result, and applies a deterministic safety/repair layer. In a 150-example non-oracle LLM run, Critical Span Guard achieves zero measured direct identifier leaks, lower quasi-identifier risk (0.127 versus 1.733 for generic prompting and 0.573 for privacy-first prompting), and higher audited task-critical fact retention (0.994 versus 0.938 and 0.529). A handwritten stress slice further shows why the problem is not reducible to “redact more”: some sensitive facts, such as pregnancy, sexual orientation, domestic violence, or minority status, can also be the task-critical answer. The results support evaluating de-identification systems by both privacy protection and retained domain meaning.

1 Introduction

Sensitive clinical and legal text is valuable for retrieval-augmented generation, supervised fine-tuning, model evaluation, safety auditing, and synthetic-data generation. The usual first step before such use is de-identification, anonymization, or pseudonymization. However, a system optimized only for removing personally identifying information can destroy the very facts that make the text useful. In a clinical note, replacing “methotrexate 15 mg weekly” with “a medication” may preserve broad topical similarity while breaking the answer to a treatment question. In a legal excerpt, replacing “Article 6” with “a specified article” removes the legal basis needed to reason about the claim. These are not cosmetic changes: they alter the information that downstream models or evaluators are expected to use.

This tension is especially important for foundation-model data pipelines. A release pipeline that over-redacts data may be privacy-conservative but unable to support clinical decision-support evaluation, legal QA, or domain-specific retrieval. A pipeline that preserves all clinically or legally meaningful facts may leave quasi-identifiers, rare events, or sensitive attributes that can aid re-identification. In practice, data stewards must make policy choices about which facts are safe to preserve, which facts must be generalized, and which facts are too sensitive to release even when they are task-relevant. Yet many evaluations still reduce de-identification to span-level recall over predefined identifier categories.

We ask a narrower, practical question: *after anonymization, is the transformed text still useful for the domain task it was meant to support?* We focus on two domains where the privacy-utility tension is concrete. Clinical-style vignettes contain diagnoses, medications, doses,

44 labs, symptoms, procedures, and timelines. Public legal snippets derived from the Text
 45 Anonymization Benchmark (TAB) contain application identifiers, party roles, claims, legal
 46 articles, procedural facts, confidential attributes, and outcomes [Pilan et al. \(2022\)](#). Existing
 47 tools such as Microsoft Presidio [Microsoft \(2026\)](#) and regulatory frameworks such as HIPAA
 48 de-identification guidance [U.S. Department of Health and Human Services \(2012\)](#) are essen-
 49 tial privacy infrastructure, but they do not by themselves answer whether the remaining
 50 text preserves task-critical meaning.

51 This paper makes four contributions. First, we construct a 150-example controlled bench-
 52 mark with synthetic clinical vignettes and TAB-derived legal snippets, each annotated with
 53 direct identifiers, quasi-identifiers, task-critical facts, and downstream QA items. Second,
 54 we propose *Task-Critical Fact Retention* (TCFR), a fact-level utility metric that distinguishes
 55 retained domain meaning from generic surface similarity. Third, we introduce Critical Span
 56 Guard, a simple anonymization protocol that extracts privacy spans and task-critical facts,
 57 anonymizes under explicit preservation constraints, and applies deterministic safety/repair
 58 rules for direct identifiers and known domain-specific failure modes. Fourth, we provide a
 59 bounded empirical audit showing that generic prompting tends to preserve utility but leak
 60 identifiers, privacy-first prompting removes identifiers but often removes the answer, and
 61 task-aware structure improves the observed privacy–utility frontier.

62 Our goal is not to claim formal anonymization, regulatory compliance, or broad clinical
 63 deployment. Instead, we argue that span-level privacy evaluation alone can both miss
 64 residual risk and reward over-redaction. For sensitive text used in foundation-model
 65 workflows, a more useful evaluation must ask two questions jointly: what identifying
 66 information remains, and what domain meaning has been lost?

67 2 Related Work

68 **Text de-identification.** Clinical de-identification has a long history in shared tasks and
 69 patient-note benchmarks. Early neural systems framed the problem as sequence label-
 70 ing over protected-health-information spans [Dernoncourt et al. \(2016\)](#), while annotation
 71 resources from i2b2 and related challenges supported systematic PHI recognition and nor-
 72 malization [Stubbs et al. \(2015\)](#). Recent work has explored stronger neural and LLM-based
 73 de-identification, but the dominant evaluation mode is still identifier detection or span
 74 replacement. This is a necessary component of privacy evaluation, but it does not directly
 75 measure whether clinically material facts such as drugs, doses, lab values, and procedures
 76 remain intact.

77 **Legal anonymization.** TAB provides an important benchmark for legal-text anonymiza-
 78 tion by annotating European Court of Human Rights cases with entity spans, masking
 79 decisions, confidential attributes, and coreference relations [Pilan et al. \(2022\)](#). Later work on
 80 neural legal anonymization further shows that legal documents require more than generic
 81 named-entity recognition because case identifiers, party roles, procedural details, and confi-
 82 dential attributes interact with legal meaning [Papadopoulou et al. \(2023\)](#). Our legal split
 83 builds on TAB but adds a utility target: legal article references, claims, outcomes, and
 84 procedural facts should be preserved when they are necessary for downstream reasoning.

85 **Privacy risk beyond direct identifiers.** Classical privacy work distinguishes direct identi-
 86 fiers from quasi-identifiers: attributes that may not identify a person alone but can do so in
 87 combination [Sweeney \(2002\)](#); [Narayanan & Shmatikov \(2008\)](#). Differential privacy gives a
 88 formal framework for aggregate releases and learning algorithms [Dwork et al. \(2006\)](#), but
 89 most practical text anonymization pipelines do not provide such guarantees. Recent LLM
 90 privacy work shows that powerful models can infer personal attributes from seemingly
 91 innocuous text [Staab et al. \(2023\)](#). Recent anonymization benchmarks therefore increasingly
 92 consider residual re-identification risk and utility together [Loiseau et al. \(2025\)](#); [Krčo et al. \(2026\)](#). We share this motivation, but make the utility target explicit at the level of domain
 93 facts and QA answers.

95 **Utility and faithfulness metrics.** Generic text-generation metrics such as token overlap,
 96 embedding similarity, or BERTScore Zhang et al. (2020) can be useful diagnostics, but they
 97 do not encode what domain facts must be preserved. High similarity to the original can
 98 even be a warning sign in anonymization because identifiers and quasi-identifiers may
 99 remain. Conversely, low similarity may be acceptable if the transformed text preserves the
 100 clinically or legally relevant answer. This motivates a fact-level metric rather than relying
 101 only on surface similarity.

102 3 Benchmark and Task

103 **Task definition.** Given an input text x , a de-identification method outputs transformed text
 104 $\hat{x}$. Each example contains three types of annotation: direct identifiers D , quasi-identifiers Q ,
 105 and task-critical facts F . Direct identifiers include names, contact information, exact dates,
 106 medical record numbers, legal application numbers, and named institutions tied to a per-
 107 son. Quasi-identifiers include age, occupation, city, nationality, sensitive attributes, family
 108 relations, rare events, and combinations of details that may enable linkage. Task-critical
 109 facts include information needed to answer a domain question: clinical diagnoses, labs,
 110 medications, doses, symptoms, procedures, timelines, or legal articles, claims, procedures,
 111 outcomes, and reasoning. A good output should remove or generalize D and high-risk Q
 112 while preserving the facts in F at the specificity required by the downstream task.

113 **Clinical split.** We generate 75 synthetic clinical vignettes from templates. Each example
 114 contains direct identifiers such as patient name, clinician name, clinic, date, medical record
 115 number, phone number, and email; quasi-identifiers such as age, city, and occupation; and
 116 task-critical facts such as diagnosis, medication, dose, lab result, symptom duration, imaging
 117 result, or treatment plan. The clinical split is synthetic by design: it provides controlled
 118 utility labels without exposing real protected health information. It should be interpreted as
 119 a controlled stress test, not as evidence of performance on real clinical notes.

120 **Legal split.** We sample 75 short excerpts from TAB, an open-source corpus of English-
 121 language European Court of Human Rights cases annotated with personal identifiers,
 122 masking decisions, confidential attributes, and coreference relations Pilon et al. (2022). For
 123 each sampled snippet, we keep short procedure, complaint, and outcome sentences when
 124 available. We use TAB spans for privacy labels and heuristic extraction plus manual spot
 125 checks for legal articles, claims, outcomes, and QA answers. This lets us study legal meaning
 126 without processing private legal files.

127 **Stress slice.** We also construct a 12-example handwritten diagnostic slice that stresses
 128 cases where privacy and utility overlap: pregnancy, sexual orientation, domestic violence,
 129 Roma marriage, exact ages, rare occupations, case labels, secret evidence, exact legal arti-
 130 cle/protocol specificity, numeric labs, and dose preservation. This slice is not used as the
 131 headline non-oracle result; it is an interpretability check on the privacy–utility frontier.

132 4 Metrics

133 We report privacy and utility metrics together. *Direct leak* is the fraction of rows where any
 134 gold direct identifier remains in the anonymized text. *QI risk* is the mean quasi-identifier
 135 risk score, where retained specific dates, locations, occupations, sensitive attributes, or
 136 combinations of details increase the risk. We also report direct, quasi, and all-span privacy
 137 recall: the fraction of gold privacy spans not found verbatim after anonymization.

138 For utility, exact TCFR measures the fraction of gold task-critical facts retained by determin-
 139 istic phrase or alias matching:

$$\text{TCFR}(\hat{x}) = \frac{1}{|F|} \sum_{f \in F} \mathbb{1}[f \text{ is retained in } \hat{x}]. \quad (1)$$

Exact matching is reproducible but conservative because faithful paraphrases may not match the gold string. We therefore also report audited TCFR using a structured LLM fact judge that marks each fact as preserved, generalized-but-acceptable, omitted, contradicted, or unclear. The judge is strict for medication names, doses, lab values, legal Article numbers, outcomes, and legal claims. *QA consistency* asks whether downstream questions can still recover the gold answers from the anonymized text. Finally, we report surface metrics such as token F1, TF-IDF cosine, and edit rate only as diagnostic foils, not as privacy or task-utility guarantees.

5 Methods

We compare three non-oracle LLM anonymizers in the 150-example OpenAI run. All methods operate on the same inputs and are evaluated with the same gold privacy and utility annotations.

Generic LLM. The generic prompt asks the model to remove personally identifying information while preserving document meaning. It resembles the simple instruction a data steward might give to a general-purpose model. This baseline tests whether broad de-identification instructions are sufficient.

Privacy-first LLM. The privacy-first prompt asks the model to remove or generalize any information that could identify a person, including rare locations, exact dates, named institutions, rare occupations, unique events, and combinations of details. It is intentionally conservative. This baseline tests the cost of resolving uncertainty by redacting or generalizing aggressively.

Critical Span Guard. Critical Span Guard uses the same low-cost model family but adds structure. First, an extraction prompt identifies privacy spans and task-critical domain facts. Second, an anonymization prompt receives these structures and is instructed to replace direct identifiers with typed placeholders, generalize quasi-identifiers when possible, preserve task-critical facts, and use the least identifying version of facts that are both privacy-relevant and task-critical. For clinical text, the prompt explicitly protects medications, doses, labs, diagnoses, symptoms, imaging findings, and meaningful timelines. For legal text, it protects legal article/statute numbers, claims, procedural posture, outcomes, and reasoning, while treating legal case/application numbers as direct identifiers. Third, a verification/repair prompt checks for leaked direct identifiers, generalizable quasi-identifiers, omitted facts, contradictions, and invented facts. Finally, a deterministic safety/repair layer replaces direct identifiers such as case/application numbers, dates, MRNs, phone numbers, emails, names, and institutions, and repairs known domain mistakes such as legal Article over-generalization and medication placeholdering.

Diagnostic local baselines. We also run local diagnostic baselines on the original benchmark, a 200-example local expansion, and the stress slice: regex rules, Presidio, direct-span oracle replacement, privacy-first oracle redaction, and a CSG oracle using gold benchmark annotations. These oracle variants are not deployable systems; they diagnose upper bounds and failure modes.

6 Experimental Setup

The promoted non-oracle run contains 150 examples: 75 synthetic clinical vignettes and 75 TAB-derived legal snippets. The LLM methods are run with a low-cost OpenAI model and cached responses. The exact model is not central to our claim: the paper evaluates prompt protocols and safety layers under a bounded budget rather than training a new model. We use no real protected health information. Clinical examples are synthetic, and legal examples are public TAB-derived snippets.

Method	Direct leak ↓	QI risk ↓	Exact TCFR ↑	Audited TCFR ↑	QA ↑
Generic LLM	0.180	1.733	0.813	0.938	0.888
Privacy-first LLM	0.000	0.573	0.398	0.529	0.414
Critical Span Guard	0.000	0.127	0.884	0.994	0.974

Table 1: Main 150-example non-oracle results. Generic prompting preserves many facts but leaks identifiers; privacy-first prompting protects direct identifiers but often removes the answer; Critical Span Guard improves the measured privacy–utility frontier.

Domain	Method	Direct leak	QI risk	Exact TCFR	Audited TCFR	QA
Clinical	Generic	0.027	2.027	0.704	0.933	0.853
Clinical	Privacy-first	0.000	1.067	0.193	0.458	0.227
Clinical	Critical Span Guard	0.000	0.000	0.820	0.996	1.000
Legal	Generic	0.333	1.440	0.922	0.942	0.922
Legal	Privacy-first	0.000	0.080	0.602	0.600	0.602
Legal	Critical Span Guard	0.000	0.253	0.949	0.993	0.949

Table 2: Domain split for the 150-example non-oracle run. The legal split exposes the hardest residual tradeoff: privacy-first redaction attains lower QI risk by deleting or generalizing legally meaningful details, while Critical Span Guard preserves more legal utility but retains a small number of task-overlapping quasi-identifiers.

We compute deterministic privacy and exact-utility metrics from gold spans and facts. We then run an LLM-assisted fact-retention audit over the anonymized outputs. Bootstrap confidence intervals and paired deltas are computed over the same examples to compare Critical Span Guard against each baseline. We also inspect qualitative failures and maintain explicit claim boundaries: the privacy metrics are string/span and heuristic risk audits, not k-anonymity, differential privacy, legal compliance, or protection against an adversary with external databases.

7 Results

Table 1 shows the core privacy–utility tradeoff. Generic prompting preserves many task facts, but leaks direct identifiers in 27 of 150 examples and has high QI risk. Privacy-first prompting removes all measured direct identifiers, but exact TCFR falls to 0.398 and QA consistency to 0.414. Critical Span Guard achieves zero measured direct leaks, the lowest overall QI risk, audited TCFR of 0.994, and QA consistency of 0.974.

The domain split in Table 2 reveals different failure modes. In clinical examples, generic prompting leaves nearly all quasi-identifiers, whereas privacy-first prompting often replaces specific diagnoses, drugs, labs, or doses with vague placeholders. Critical Span Guard removes all measured clinical QI hits while retaining almost all audited facts. In legal examples, the frontier is harder. Privacy-first prompting has lower QI risk than Critical Span Guard on the legal subset because it generalizes legal claims, jurisdictions, statutes, or sensitive attributes. This improves the measured privacy score but damages the legal utility target.

The span-level privacy audit agrees with the row-level picture. Critical Span Guard reaches direct span recall of 1.000 and all privacy-span recall of 0.982, compared with 0.919 and 0.771 for generic prompting and 1.000 and 0.964 for privacy-first prompting. Paired deltas on the same examples sharpen this interpretation. Relative to generic prompting, Critical Span Guard reduces direct-leak rate by 0.180 (95% bootstrap CI 0.120–0.247) and QI risk by 1.607 (1.407–1.807). Relative to privacy-first prompting, it improves exact TCFR by 0.487 (0.434–0.538), audited TCFR by 0.466 (0.416–0.514), and QA consistency by 0.560 (0.498–0.620). The legal-only QI comparison is an important caveat: Critical Span Guard is slightly worse than privacy-first prompting on legal QI risk because it intentionally preserves some legal facts that overlap with quasi-identifying content.

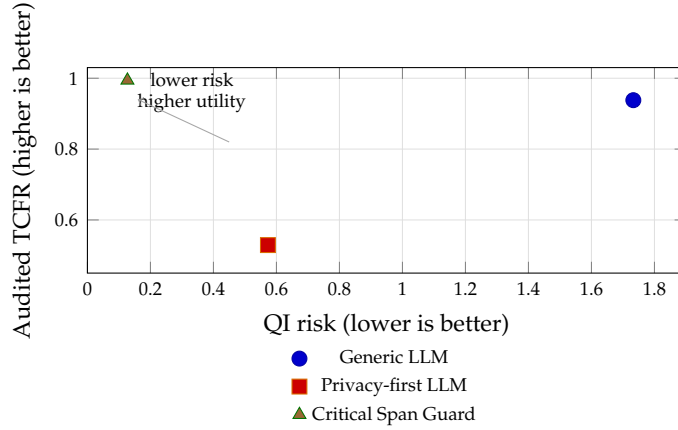


Figure 1: Privacy-utility frontier for the 150-example non-oracle LLM run. Critical Span Guard is closest to the upper-left target region: lower measured quasi-identifier risk and higher audited task-critical fact retention.

Stage	Direct leak	QI risk	Exact TCFR	QA
CSG draft	0.073	1.027	0.881	0.969
CSG verified	0.073	1.027	0.881	0.969
CSG + safety/repair	0.000	0.127	0.884	0.974

Table 3: Critical Span Guard ablation reconstructed from cached outputs. The verification prompt did not change deterministic metrics in this sample; the measured gain comes from the deterministic safety/repair layer.

To test judge dependence, we also ran an external GPT-5.5 fact-retention audit on a stratified 60-example subset containing all residual CSG QI rows, all strict CSG audited-loss rows, and additional high-stress cases. The stronger judge is stricter than the original low-cost judge but preserves the ranking: audited TCFR is 0.917 for Critical Span Guard, 0.875 for generic prompting, and 0.483 for privacy-first prompting. Fact-label agreement with the original audit is 0.912, 0.935, and 0.833 respectively, with most disagreements caused by strict clinical numeric or legal specificity. A GPT-5.5 adversarial privacy audit on the 20 hardest rows is similarly conservative: generic prompting has direct privacy flags on 60% of rows, Critical Span Guard on 5%, and privacy-first prompting on 0%, but privacy-first is the same method that loses most task-critical facts. We treat these external audits as robustness checks, not as expert annotation.

Figure 1 summarizes the main privacy-utility frontier: the task-aware protocol moves toward lower QI risk without collapsing audited fact retention.

Table 3 is important for claim discipline. In this cached sample, the verification/repair prompt did not change deterministic metrics: the draft and verified rows are numerically identical. The measured gain comes from the deterministic safety/repair layer, which closes eleven direct leaks, reduces QI-risk rows from 74 to 11, repairs legal Article specificity losses, and leaves eleven legal residual cases. This result suggests that LLM-generated anonymization should be paired with explicit deterministic checks, not trusted as a single-pass transformation.

Table 4 addresses whether generic utility metrics are enough. Generic prompting and Critical Span Guard have nearly identical token F1 against the original text, and generic prompting has higher TF-IDF cosine similarity, but it also has far higher privacy failure and audited fact failure rates. Among the 225 method-example rows at or above the median TF-IDF cosine score, 73 still have privacy failures and 34 still have audited fact loss. Surface similarity therefore rewards text that may be too close to the original and misses task-critical

Method	Token F1	TF-IDF	Edit rate	Audited fact-fail rate	Privacy-fail rate
Generic LLM	0.776	0.708	0.247	0.240	0.747
Privacy-first LLM	0.577	0.472	0.464	0.793	0.273
Critical Span Guard	0.773	0.617	0.274	0.020	0.027

Table 4: Surface similarity metrics are not sufficient proxies for privacy-safe utility.

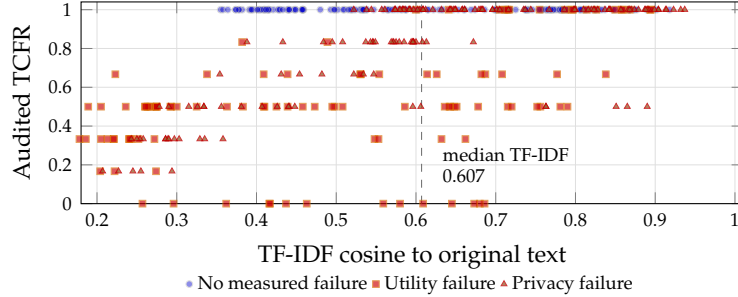


Figure 2: Surface similarity failure scatter for the 150-example non-oracle run. Among the 225 rows at or above the median TF-IDF cosine, 73 still have privacy failures and 34 have audited fact loss.

failures; this is consistent with broader concerns that generic text-generation metrics do not encode domain-specific faithfulness by themselves [Zhang et al. \(2020\)](#).

The n200 diagnostic expansion in Table 5 gives the same qualitative frontier. Regex and direct-span replacement preserve exact utility but leave high QI risk. Presidio is useful infrastructure but performs poorly on our legal snippets because the benchmark contains legal identifiers and confidential attributes that are not standard clinical or generic PII. The privacy-first oracle nearly eliminates measured QI risk but loses more than half of task-critical facts. The CSG oracle indicates that a gold-informed version of task-aware anonymization can preserve utility while substantially reducing QI risk, though it remains an upper-bound diagnostic rather than a deployable system. The domain split explains the overall Presidio row: Presidio’s legal direct-leak rate is 0.980, while its clinical TCFR is only 0.445 under exact matching.

The stress slice in Table 6 clarifies the frontier. Privacy-first oracle redaction drives measured direct leaks and QI risk to zero, but fails QA on 11 of 12 rows. Direct-span replacement preserves utility but leaves quasi-identifiers in every row. CSG oracle replacement preserves all task facts with zero direct leaks, but retains task-critical quasi-identifiers in 6 of 12 rows. This residual risk is not merely an implementation bug: some facts, such as pregnancy, sexual orientation, domestic violence, or minority status, can be both sensitive and necessary to answer the domain question.

8 Qualitative Analysis

In `clinical_0006`, the original vignette describes rheumatoid arthritis with MCP joint swelling, positive anti-CCP, and methotrexate 15 mg weekly plus folic acid 1 mg daily. The privacy-first output replaces these with “finger joint areas,” “a specific antibody test,” “an autoimmune joint condition,” and “a disease-modifying medication,” so both clinical QA questions fail. Critical Span Guard redacts the patient, clinic, date, MRN, phone, email, clinician, city, and exact age while preserving the lab, diagnosis, medication, and doses.

In `legal_0006`, generic prompting leaves application number 37645/97 and the exact filing date while preserving Article 6. The privacy-first output removes those identifiers but replaces Article 6 with “a specified article,” breaking the legal-basis question. Critical Span

Method	Direct leak	QI risk	TCFR	QA
Regex	0.075	1.780	0.993	0.993
Presidio	0.490	2.490	0.678	0.786
Direct-span oracle	0.000	2.810	1.000	1.000
Privacy-first oracle	0.000	0.005	0.448	0.336
CSG oracle	0.000	0.225	1.000	1.000

Table 5: Local n200 diagnostic expansion. These local/oracle baselines are not the headline non-oracle result; they test whether the qualitative frontier persists under a larger deterministic diagnostic slice.

Method	Direct leak	QI risk	Exact TCFR	QA
Regex	0.083	2.167	1.000	1.000
Presidio	0.667	2.667	0.817	0.833
Direct-span oracle	0.000	2.500	1.000	1.000
Privacy-first oracle	0.000	0.000	0.608	0.528
CSG oracle	0.000	0.667	1.000	1.000

Table 6: Handwritten robustness stress slice. This diagnostic slice is separate from the non-oracle headline run.

Guard removes the application number, name, country, institution, nationality, and date while preserving Article 6 and the reasonable-time civil-proceedings claim.

The remaining frontier is visible in `legal_0005`: generic and privacy-first prompting avoid the measured QI hits by generalizing away the specific section 42 Fair Employment claim, while Critical Span Guard preserves Article 6 and the exact claim. The preserved statute name still contains “Northern Ireland” and “1976,” so the row remains a measured QI-risk case. This is a useful limitation: some legal statute, jurisdiction, regime, or offense specificity may be both important for legal reasoning and quasi-identifying. A residual audit of the eleven remaining CSG QI rows found six clear task-critical legal-overlap cases and five additional frontier candidates.

These examples motivate a policy distinction that span-level metrics cannot make alone. Some outputs are clearly bad because they leak direct identifiers. Some are clearly bad because they remove the answer. Others are frontier cases where the correct decision depends on the intended release setting. A private internal evaluation corpus may preserve more legal specificity than a public release. A public synthetic-data release may need to generalize facts that are otherwise useful. Our metric does not resolve these policy choices, but it makes them visible.

9 Limitations and Ethics

This is a small fast-iteration study, not a formal anonymization guarantee. The clinical split is synthetic and should not be presented as evidence on real clinical notes. The legal split is public TAB-derived text, but legal fact and claim extraction is heuristic and manually spot-checked rather than fully expert annotated. Audited TCFR uses an LLM judge; it is useful for separating paraphrase from omission, but manual review remains necessary for final paper claims. The privacy metrics detect known direct spans and heuristic quasi-identifiers; they do not provide k-anonymity, differential privacy, legal compliance, or protection against a motivated adversary with external knowledge.

The remaining CSG quasi-identifier rows should be read as audit targets, not as solved cases: several involve quasi-identifying text inside the annotated legal claim, and five are frontier candidates requiring policy review. A more privacy-aggressive release policy could generalize those claims, but that would change the downstream legal utility target. In the promoted 150-example run, three CSG rows have audited fact loss under strict labeling: `legal_0016` generalizes “Polish authorities” to “authorities,” while `clinical_0055` and

clinical_0059 generalize or omit strict clinical specificity. These rows mark jurisdictional, temporal, and diagnostic specificity as policy-sensitive utility targets rather than simple privacy failures.

Threat model. We evaluate direct string leakage and heuristic quasi-identifier retention under a bounded controlled-release scenario: an inspector can search transformed text for known direct identifiers and retained quasi-identifying details. We do not model linkage attacks by an adversary with external databases or case-specific background knowledge. Public redistribution should therefore preserve provenance and license information and include residual-risk notes, especially for TAB-derived legal snippets.

The intended use of this work is evaluation and safer data transformation research. We do not release or process real protected health information. The main ethical lesson is that aggressive redaction can make sensitive data safer to share only by making it useless or misleading. Responsible release pipelines should therefore state the downstream utility requirements explicitly and audit both residual privacy risk and task-critical meaning loss.

10 Conclusion

De-identification systems should be evaluated not only by what they remove, but also by whether they preserve the facts needed for downstream reasoning. On a controlled clinical/legal benchmark, generic anonymization leaks identifiers and quasi-identifiers, privacy-first anonymization often removes the answer, and a small amount of task-aware structure improves the observed privacy–utility tradeoff. The stress slice further shows that some privacy–utility conflicts are inherent: the sensitive fact may also be the task-critical fact. This makes task-critical utility evaluation a necessary complement to span-level privacy metrics for sensitive text transformation.

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